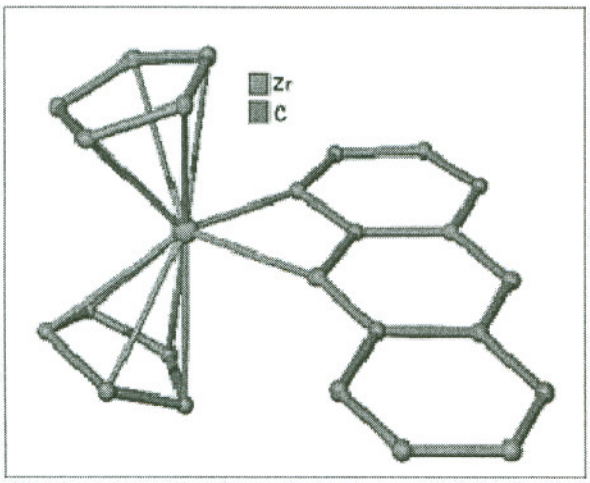
Inorganic Cumulative Exam

November 10, 2006

The following was copied from the webpage of our most recent general seminar speaker.

Organometallic Chemistry on the Edge (of Polycyclic Aromatic Carbon Compounds)

In another project, we are preparing and studying complexes with a metal or metals bonded to the edge of polycyclic aromatic compounds. An example of a zirconium complex is shown in the figure on the left. Polycyclic aromatic carbon compounds and their derivatives are useful in a wide variety of applications in electronics, optics, and other materials areas. [redacted]



1. What was the speaker's name? (Hint: His webpage lists his first name as P-Au-L)

2. Apply the "18 electron" counting conventions to the zirconium compound pictured. What is the electron count and how does this fit into the general "18 electron rule"?

3. The author focused his talk on a specific application and this application was ultimately used to justify funding. What was this application and how did 5-membered rings fit into the talk?

4. How would you justify the study of boron nitrides in this context?

5. If you were working on this project what **specific** health related concerns would you have?